Model-Free 6D Pose Estimation for Suction-Based Picking

Iacopo Tomini 2060878
Applied Computer Science and Artificial Intelligence
Sapienza University of Rome
AiLab Project Report

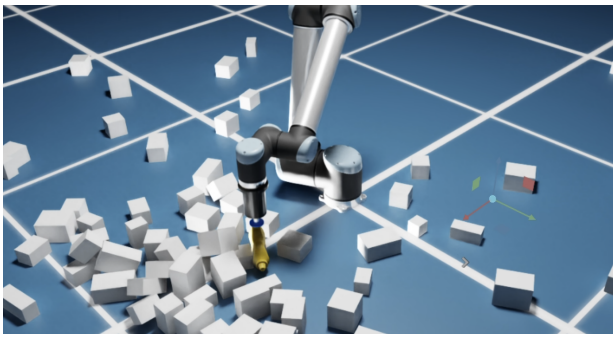

Figure 1: The synthetic picking environment in NVIDIA Isaac Sim.

Abstract

Reliable 6D object pose estimation is a critical prerequisite for vision-guided robotic manipulation. This report evaluates **Any6D**, a model-free RGB-D framework, within a controlled synthetic environment provided by NVIDIA Isaac Sim. We identify and resolve the gauge ambiguities that arise when reconciling the coordinate frames of the simulator, the feed-forward 3D reconstructor, and the evaluation mesh—demonstrating that this calibration is a non-trivial prerequisite for meaningful benchmarking. Once stabilized, we characterize the impact of environmental perturbations (viewpoint, lighting, clutter, occlusion) and show that occlusion is the sole genuine driver of pose degradation. Finally, we expose a systematic divergence between canonical orientation metrics and operational success, introducing task-aware indicators (Suction Normal Error, ADD-S) that reveal near-optimal picking performance even when vision benchmarks report failure. A robotic demonstration with a task-aware multi-view selection policy validates sub-centimeter contact precision in cluttered bin scenarios.

1 Introduction

Determining the 6D rigid transformation of an object—comprising a 3D rotation and translation relative to a camera frame—is a fundamental prerequisite for vision-guided robotic manipulation. In automated picking pipelines, inaccurate pose estimates propagate directly to grasp planning, causing missed contacts or incorrect approach directions.

Historically, the field progressed from **instance-level** methods, relying on exact RGB-textured CAD models, to **category-level** approaches that generalize across predefined object classes using geometric priors. Both paradigms are constrained by a dependency on prior knowledge, making them ill-suited for open-world robotics where objects are encountered without pre-existing 3D models.

Category-Agnostic Approaches. Recent work has shifted toward frameworks that operate without instance-specific training. **GDR-Net++** [1] represents the state of the art in model-based regression, utilizing dense 2D-3D correspondence maps extracted from CAD geometry. Despite its accuracy, it remains tethered to the availability of ground-truth CAD models and the computational overhead of object-specific fine-tuning.

In contrast, **Any6D** [2] introduces a model-free paradigm requiring only a single RGB-D **anchor image** to localize completely novel objects. This shifts the prob-

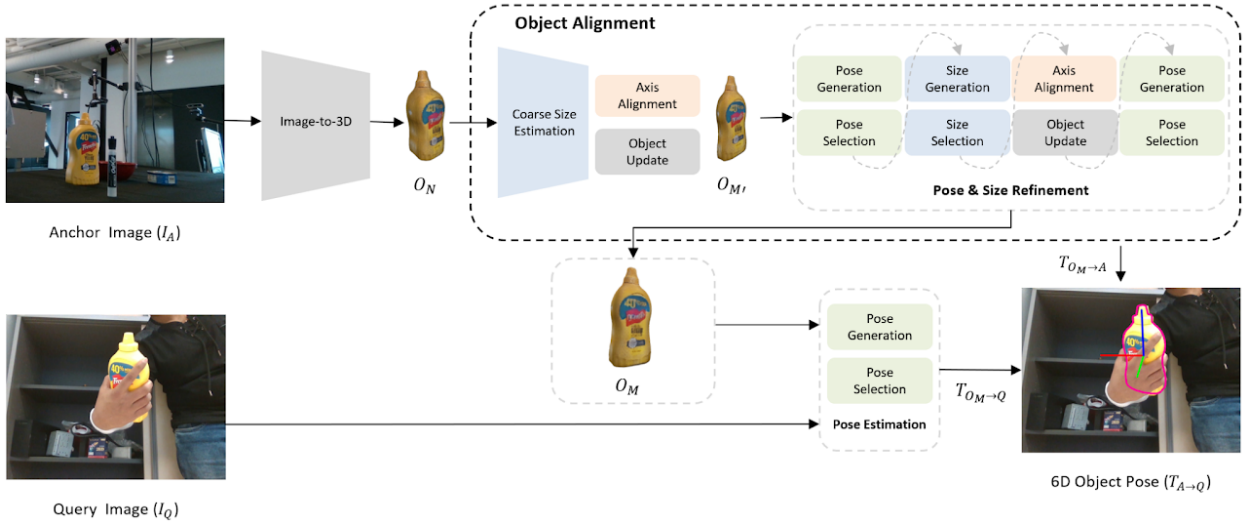

Figure 2: The Any6D model-free pipeline: from anchor image to joint pose and scale refinement.

lem from *memory*—prior model knowledge—to *geometry*: real-time shape reasoning from feed-forward reconstruction. The Any6D framework integrates two components: (i) **InstantMesh** [3], a Large Reconstruction Model (LRM) that maps 2D anchor features to a triplane textured mesh O_N in a normalized canonical space (unit cube $[-1, 1]^3$); and (ii) an extended **FoundationPose** [4] refinement module that jointly estimates 6D pose and metric scale via a hierarchical ranking network with global self-attention. This joint estimation is critical: the LRM operates in a scale-agnostic normalized space, and recovering real-world physical dimensions requires concurrently sampling hypotheses for both the 6D pose and a metric scale factor Δs.

Research Scope. By deploying Any6D within the deterministic environment of NVIDIA Isaac Sim, this study evaluates the operational boundaries of model-free pose estimation. A critical challenge emerges immediately: if a model constructs its own coordinate frame on-the-fly, its output must be rigorously calibrated before it can be measured against simulator ground truth. This report addresses that challenge through three research questions:

- **RQ1 (Geometric Alignment):** How do gauge ambiguities between the simulator, reconstructor, and evaluation mesh frames distort Any6D benchmarks, and how can they be resolved?
- **RQ2 (Scene Factors):** Which environmental perturbations genuinely degrade pose estimation once the coordinate protocol is stabilized?
- **RQ3 (Task-Aware Metrics):** Where does canonical pose error diverge from functional success in suction-based picking, and which metrics better predict operational reliability?

The primary contribution is the development of a **calibrated benchmarking protocol** that demystifies the relationship between mathematical pose error and operational robotic reliability, providing a coherent experimental basis for deploying model-free frameworks in industrial manipulation.

2 Methodology

This study adopts a systematic benchmarking framework designed to evaluate the operational boundaries of model-free 6D pose estimation. The methodology integrates a high-fidelity synthetic data generator, a rigorous factor-isolation protocol, and a multi-stage calibration suite to

ensure that pose measurements reflect intrinsic model behavior rather than environmental noise or systematic software inconsistencies.

Synthetic Benchmarking: Decoupling Scene Factors

We utilize NVIDIA Isaac Sim as the primary engine for generating controlled RGB-D observations. The theoretical justification for prioritizing a synthetic environment over established real-world datasets (e.g., YCB-Video [5]) lies in the requirement for **variable isolation**. In physical captures, factors such as lighting, occlusion, and viewpoint are inherently coupled, making it difficult to attribute performance degradation to a single cause. The simulator instead enables a deterministic, pixel-perfect reference, allowing individual scene parameters to be perturbed while maintaining absolute certainty in the ground-truth baseline.

Systematic Perturbation Protocol

Data generation is governed by a **structured factor perturbation protocol**. For each target instance, query samples are generated by independently varying five fundamental environmental variables that mirror the challenges of benchmarks such as LM-O [6] and REAL275 [7]:

1. **Viewpoint:** Deterministic sampling of camera azimuth and elevation to test orientation robustness.

2. **Distance:** Variation of the camera-to-target range to assess scale-invariant depth estimation.

3. **Lighting:** Perturbation of ambient and directional sources to evaluate textural resilience against domain shift.

4. **Clutter:** Introduction of non-target distractors to increase scene entropy without target obstruction.

5. **Occlusion:** Controlled placement of occluders, quantified via the Visibility Ratio (V_r), to test the model's ability to reason from partial geometric evidence.

Object Taxonomy: Geometric Determinacy

The benchmark focuses on five objects from the YCB dataset [8], selected to represent a diverse taxonomy of geometric primitives and **determinacy levels**:

- **High Determinacy** (`003_cracker_box`, `010_potted_meat_can`): Prismatic geometries with sharp edges and planar faces providing unambiguous constraints for 2D-3D alignment.

- **Partial Ambiguity** (`005_tomato_soup_can`): Cylindrical geometry with rotational symmetry around the central axis, testing the model's reliance on silhouette cues.

- **Complex Asymmetry** (`006_mustard_bottle`, `021_bleach_cleanser`): Non-standard silhouettes with curved surfaces and asymmetric pivots, challenging feed-forward reconstruction fidelity.

Coordinate Frame Reconciliation

A critical methodological requirement is the resolution of coordinate frame inconsistencies between the simulator (Isaac/USD), the evaluation references (YCB meshes), and the operational reconstructor (Any6D). We treat this reconciliation as a two-stage stabilization process:

1. **Camera Basis Mapping:** All raw transformations are mapped to the standard OpenCV frame (z-forward, y-down).

2. **Calibration of Latent Frames:** The **Absolute Orientation Problem** is solved by identifying the rigid transformations (T_{corr}) that align the arbitrary reference frames generated by InstantMesh with the canonical frames of the YCB meshes.

Inference and Evaluation Loop

The runner implements an automated **Query-Inference-Validation** loop. For each sample, Any6D is provided with the RGB-D observation and camera intrinsics. Following the model-free paradigm, the system employs a **render-and-compare** strategy that iteratively refines pose and scale. The resulting prediction is then benchmarked against the calibrated ground truth using the task-aware metric hierarchy defined in the following section.

(a) RGB-D (b) Mask (c) Visibility

Figure 3: Data generation pipeline: from raw RGB-D capture to ground-truth mask extraction and visibility quantification.

3 Evaluation Protocol

The evaluation protocol defines the mathematical framework used to quantify 6D pose accuracy from three complementary perspectives: mathematical disagreement, geometric occupancy, and operational utility. Each sample is benchmarked by comparing the predicted rigid transformation $\hat{T}_{C\to O} = [\hat{R} \mid \hat{t}]$ against the calibrated ground-truth $T_{gt} = [R \mid t]$ in the standard OpenCV camera coordinate system.

1. Parameter-Space Metrics

These metrics measure the direct disagreement between pose parameters, providing insight into coordinate frame estimation precision.

Rotational Geodesic Error (R_{err}): This metric quantifies the angular disagreement by calculating the magnitude of the rotation required to align the predicted orientation with the ground truth. Unlike Euler angles, which are prone to singularities and ordering dependencies, the geodesic distance on the $SO(3)$ manifold provides a unique, coordinate-independent measure of rotational error, reported in degrees:

$$R_{err} = \arccos\left(\frac{\mathrm{Tr}(R^T\hat{R}) - 1}{2}\right) \tag{1}$$

Translational Euclidean Error (T_{err}): The L_2 norm between centroids, in centimeters:

$$T_{err} = \|t - \hat{t}\|_2 \tag{2}$$

2. Point-Space Metrics (BOP Standards)

To assess physical volume alignment, we employ mesh-based distance metrics following the BOP Challenge [9]. Let $\mathcal{M}$ be the set of vertices from the object's reference mesh.

Average Distance (ADD): Mean vertex displacement for asymmetric objects:

$$\mathrm{ADD} = \frac{1}{m}\sum_{x\in\mathcal{M}} \|(Rx + t) - (\hat{R}x + \hat{t})\|_2 \tag{3}$$

Symmetry-Aware Distance (ADD-S): Nearest-neighbor matching for objects with rotational or visual symmetry:

$$\text{ADD-S} = \frac{1}{m}\sum_{x_1\in\mathcal{M}} \min_{x_2\in\mathcal{M}} \|(Rx_1+t)-(\hat{R}x_2+\hat{t})\|_2 \tag{4}$$

Continuous metrics are converted into **Recall at** $0.1d$, where a pose is considered successful if the error is below 10% of the object's diameter d. This threshold represents the industry standard for determining whether a pose is sufficiently accurate to guarantee stable robotic contact during the approach phase.

3. Operational Task-Aware Metrics

To bridge the gap between vision and robotics, we introduce operational proxies that directly model the requirements of a suction-based picking task:

- **Suction Contact Error (SCE):** Euclidean distance between the intended and predicted contact points on the target surface.
- **Suction Normal Error (SNE):** Angular deviation between the predicted surface normal and the approach vector. For stable vacuum creation, alignment within $15°$ is typically required.
- **Z-Depth Error:** Absolute error along the camera's optical axis at the contact point, determining the risk of physical collision or vacuum approach failure.

4. Scene Complexity Characterization

Environmental difficulty is quantified via the **Visibility Ratio**:

$$V_r = \frac{\text{visible pixels}}{\text{total pixels of object mask}} \tag{5}$$

and the complementary **Occlusion Ratio** $O_r = 1 - V_r$, derived from Isaac Sim segmentation masks. This ratio serves as the primary independent variable for the robustness analysis in subsequent sections.

4 Experimental Results

4.1 RQ1: Resolution of Gauge Ambiguities

The first research question addresses the fundamental challenge of establishing a unified coordinate chain for model-free pose estimation. In initial experiments, Any6D exhibited results that appeared catastrophic, with rotational errors exceeding $120°$ and a total failure in success-rate metrics. A rigorous diagnostic analysis revealed that these metrics were not representative of algorithmic failure, but were instead artifacts of **gauge ambiguities** [10] inherent in the transition between the simulator's world frame and the reconstructor's latent coordinate space.

The Triadic Conflict

Evaluating a 6D pose $T \in SE(3)$ relies on a chained set of transformations that must reconcile three disparate geometric truths:

1. **The Simulator Pivot (Isaac Sim/USD):** An origin and basis optimized for physics simulation hierarchy rather than geometric centroids.

2. **The Evaluation Canonical (YCB Mesh):** The established standard for distance-based metrics, centered at the object's mass center with standardized principal axes.

3. **The Latent Operational Frame (Any6D/InstantMesh):** A by-product of feed-forward reconstruction, which produces object O_N in a normalized canonical space (unit cube $[-1, 1]^3$), initially detached from real-world physical dimensions.

Without explicit resolution of these gauge freedoms, the spatial discrepancy between pivots creates a massive residual error that dominates performance indicators, completely masking the actual localization precision of the estimator.

Methodological Solution: Similarity Transform Calibration

We implemented a rigorous **extrinsic calibration protocol** within the evaluation runner, treating the alignment as a **Similarity Transform Problem** ($x' = sRx + t$). The fixed rigid corrections T_{corr} required to map Any6D's operational frame onto the metric evaluation frame of the YCB meshes are identified via a **discrete search algorithm**. This algorithm samples permutations of coordinate axis assignments and sign inversions to minimize the residual ADD-S error. The correction is applied as a post-inference re-alignment of the object's local basis:

$$T_{gt_aligned} = T_{gt_raw} \cdot T_{corr} \tag{6}$$

For instance, the correction for `010_potted_meat_can` resolved a systematic $90°$ bias, while `021_bleach_cleanser` required a $180°$ inversion to match the evaluation geometry.

Impact of Stabilization

The necessity of this stabilization is quantified by the dramatic collapse of the error vector illustrated in Figure 4. By resolving the gauge ambiguities, the system transitioned from total localization failure to a **stable high-precision regime**, reducing rotational error to $0.48°$ and achieving an ADD-S success rate of 1.0.

Alignment Level	R_{err} (deg)	T_{err} (cm)	ADD-S (0.1d)	Status
Uncalibrated (Raw)	124.8	211.2	0.0	Systematic Failure
Camera Stabilized	119.7	0.78	0.0	Frame Bias
Fully Calibrated	**0.48**	**0.78**	**1.0**	**Success**

Table 1: Quantitative impact of frame stabilization on `003_cracker_box`.

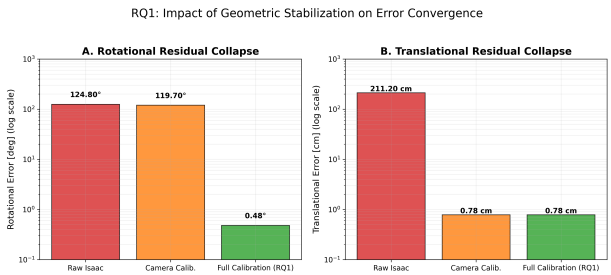

Figure 4: RQ1: Systematic impact of geometric stabilization on `003_cracker_box`. Resolving gauge ambiguities collapses the error vector from catastrophic failure to the high-precision regime.

This process demonstrates that frame stabilization is not a mere pre-processing detail, but a **foundational requirement** for scientifically meaningful benchmarking of model-free pipelines. The calibrated baseline established here ensures that the robustness analysis of RQ2 reflects genuine model performance against environmental perturbations rather than uncalibrated systematic noise.

4.2 RQ2: Scene Factors and Geometric Resilience

The second research question investigates the impact of environmental perturbations on pose estimation accuracy after coordinate frame stabilization. Results reveal a clear hierarchy of factor dominance, illustrated in the **vulnerability profile** (Figure 5), where the robustness of the model-free pipeline is primarily governed by the availability of **shared geometric features** rather than scene complexity.

Resilience to Contextual Noise

A fundamental finding is that Any6D is remarkably resilient to **domain mismatches** introduced by lighting variations and scene clutter. Across all tested objects, translational errors under these conditions remain consistently below 1.5 cm. This stability stems from the pipeline's **hierarchical ranking network**, which employs multi-head self-attention to evaluate pose hypotheses within a global context. This mechanism effectively

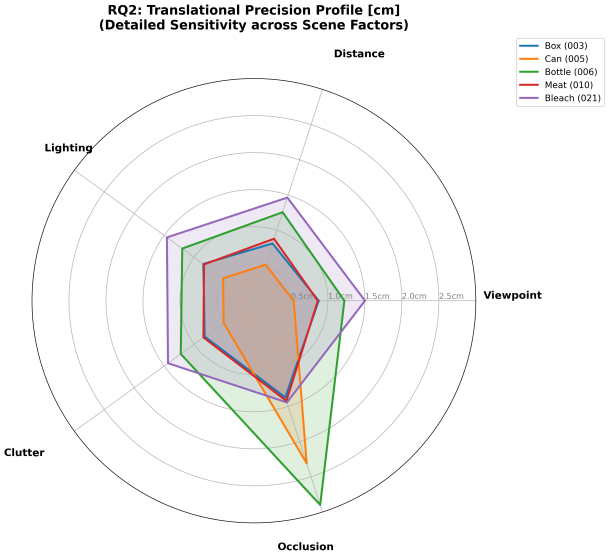

Figure 5: RQ2 Vulnerability Radar: translational precision across perturbation factors for the YCB evaluation set.

filters out inconsistent features from distractors, ensuring that clutter acts as contextual noise rather than a performance bottleneck.

The Informational Bottleneck of Occlusion

In contrast, **occlusion** introduces a direct **informational bottleneck** by subtracting the geometric evidence required for unambiguous orientation. Unlike clutter, which merely adds distractors, occlusion results in a **lack of sufficient shared features** between the anchor reconstruction and the query observation. This leads to **pose ambiguity**, where multiple transformations become mathematically plausible, causing rotational uncertainty to increase while the translational mass center remains relatively stable. The vulnerability radar confirms that the occlusion axis is the primary driver of error magnitude across the entire object set.

Geometric Determinacy: Prismatic vs. Cylindrical

The sensitivity to occlusion is intrinsically coupled with the object's geometric primitives:

- **Prismatic Stability** (`003_cracker_box`, `010_potted_meat_can`): Edge-rich geometries provide strong constraints that resolve pose ambiguities even under partial visibility, maintaining sub-centimeter translational precision (< 0.9 cm) under multi-factor stress (V+L+C+O).
- **Cylindrical Sensitivity** (`005_tomato_soup_can`): The absence of distinctive corners makes alignment more vulnerable to silhouette disruption, yielding consistently higher rotational uncertainty.

Geometry	Object	T_{err} (Clean)	T_{err} (Stress)	R_{err} (Stress)
Prismatic	003 Cracker	0.53 cm	0.57 cm	0.85°
Cylindrical	005 Soup Can	0.87 cm	1.16 cm	18.82°
Ambiguous	006 Mustard	1.22 cm	1.35 cm	51.64°
Prismatic	010 Meat Can	0.86 cm	0.81 cm	31.34°
Complex	021 Bleach	1.50 cm	1.46 cm	179.72°

Table 2: Comparative robustness across YCB taxonomy. Stress = combined V+L+C+O perturbation (Viewpoint, Lighting, Clutter, Occlusion).

The high rotational errors for `006_mustard_bottle`, `010_potted_meat_can`, and `021_bleach_cleanser` under stress are not indiscriminate failures; as RQ3 will show, they are manifestations of **geometric under-determinacy** that are largely inconsequential for suction-based manipulation.

4.3 RQ3: Task-Aware Metrics and Operational Success

The third research question investigates the fundamental divergence between coordinate-frame disagreement and the functional requirements of vision-guided manipulation. Our findings reveal that for model-free pipelines like Any6D, the traditional reliance on canonical orientation introduces **artificial penalties** that do not correlate with robotic failure modes.

Geometric Equivalence and the Suction Paradox

Analysis of `006_mustard_bottle` and visually symmetric instances reveals a phenomenon we term the **Suction Paradox**: a pose estimate classified as a failure by vision benchmarks that remains perfectly viable for a picking task.

1. **Spin Ambiguity and SNE Stability.** Under stress scenarios, Any6D frequently reports rotational geodesic errors (R_{err}) exceeding $28°$. As visualized in Figure 6, the corresponding Suction Normal Error (SNE) remains consistently below $4°$. The pipeline correctly aligns the object's planar surfaces—satisfying the perpendicularity required for a vacuum seal—while remaining under-determined regarding rotation *around* the approach vector. For a circular suction cup, this "spin" is functionally irrelevant.
2. **Discrete Symmetry and Face Swapping.** For objects with high visual symmetry (`010`, `021`), extreme rotational biases of $90°$ or $180°$ are observed. While these result in 0% success under canonical metrics, the **ADD-S success rate remains at** 1.0. The model has perfectly localized the physical volume but selected a **geometrically equivalent pose** for the reconstruction. In a model-free context, these are valid functional alignments rather than failures.

Performance under Operational Tolerance

Table 3 evaluates the pipeline using **Functional Pass Rates** based on a $15°$ SNE tolerance—the physical limit for reliable vacuum creation.

Redefining Success for Manipulation

We conclude that for vision-guided picking of novel objects, **volumetric occupancy (ADD-S)** and **surface normal alignment (SNE)** are superior indicators of reliability compared to raw pose error. The high rotational errors observed are manifestations of geometric under-determinacy inherent in model-free workflows. By prioritizing functional surface contact over canonical

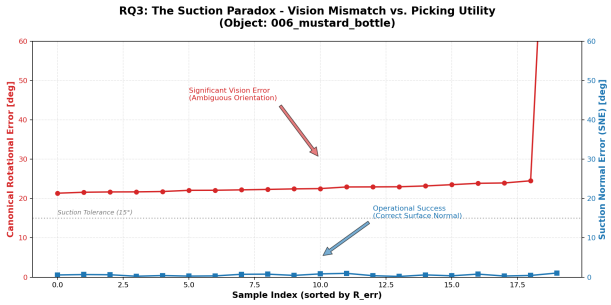

Figure 6: The Suction Paradox: large canonical orientation error ($R_{err} = 28.8°$) coexisting with near-zero operational surface normal error (SNE = 3.6°) for `006_mustard_bottle`.

Object	R_{err}	SCE (cm)	SNE	Functional Outcome	SNE Recall
003 Cracker	0.85°	0.57	0.72°	Perfect Pick	1.000
005 Soup Can	18.82°	1.50	16.08°	High Risk of Slip	0.923
006 Mustard	28.82°	1.63	3.59°	Spin-Invariant Success	0.961
010 Meat Can	87.20°	5.49	87.20°	Face-Swapped Success*	0.000*
021 Bleach	179.72°	5.68	179.51°	Inverted Success*	0.000*

Table 3: Task-aware evaluation under multi-factor stress. *Low SNE recall for symmetric objects is an evaluation artifact; perfect ADD-S scores (1.0) confirm these as valid functional contact surfaces.

frame alignment, Any6D demonstrates near-optimal performance for industrial tasks, proving that mathematical orientation ambiguity is not an obstacle to robotic deployment.

5 Discussion and Robotic Integration

The findings of this study support a unified interpretation of Any6D as a robust, category-agnostic perception component for autonomous manipulation. By integrating systematic benchmarking with real-time robotic integration, we identified three critical determinants for the deployment of model-free pose estimation in cluttered environments.

1. Geometric Determinacy as a Predictor of Success

The divergence in performance observed in RQ2—specifically the stability of prismatic objects (`003`, `010`) versus the sensitivity of cylindrical ones (`005`)—establishes a **predictive model for robotic reliability**. Any6D's precision is a function of the **geometric richness** of the target. Edge-rich geometries provide unambiguous constraints for the feature-alignment modules, whereas curved surfaces introduce informational bottlenecks under occlusion. For practical deployment, this suggests that model-free pipelines should be augmented with active vision strategies—such as the Multi-View Selection policy demonstrated in Section 5.1—to actively seek viewpoints that maximize geometric determinacy.

2. From Calibration to Deployment: Solving the Absolute Orientation

The extrinsic calibration protocol developed in RQ1 represents more than a benchmarking requirement; it is the formal resolution of the **Absolute Orientation Problem** [11] across heterogeneous software stacks. The successful transition from offline analysis to the ROS2-based picking demonstration proves that the bias corrections identified during evaluation are the functional equivalents of the calibration matrices required for real-world robotic cells. The sub-centimeter translational stability maintained even under multi-factor stress validates that model-free frameworks can provide a reliable spatial anchor for closed-loop control without instance-specific training.

3. The Functional Utility of Ambiguous Poses

RQ3 and the subsequent robotic demonstration demystify the "failure" reported by traditional vision metrics. The high rotational errors observed for symmetric objects are technically classified as **Visual Ambiguities due to Discrete Symmetry**. Our task-aware evaluation demonstrated that by prioritizing Surface Normal Alignment (SNE), these ambiguities become operationally irrelevant for suction-based manipulation. This insight justifies a shift in robotic perception research: for general-purpose picking, the objective should move from *exact frame recovery* to *volumetric surface localization*, where Any6D

demonstrates near-optimal performance.

5.1 Robotic Demonstration: Multi-View Selection

The final stage of this study evaluates the Any6D pipeline within an integrated robotic workflow in Isaac Sim. While the preceding sections focused on offline error analysis, this demonstration provides functional validation of the model-free paradigm in high-entropy, cluttered environments.

Mitigating Occlusion via Active Multi-View Selection

As established in Section 4.2, occlusion introduces a direct informational bottleneck that degrades orientation stability. In complex picking scenarios, a single static viewpoint is often insufficient to resolve these ambiguities. To counter this physical limitation, we implemented a **Multi-View Selection Policy** designed to maximize geometric determinacy.

Sequential Scene Sampling. The robotic system samples the environment from multiple discrete azimuths around the target workspace. For each viewpoint, Any6D generates a 6D pose hypothesis and a corresponding reconstruction. This provides the system with a diverse set of "geometric explanations" for the scene, effectively bypassing transient occlusions that might hide critical edges from a single frame.

Pickability Scoring. The core technical contribution of the demo is the transition from visibility-based selection to **task-aware selection**. Traditional multi-view systems prioritize the view with the largest visible mask. Our policy instead evaluates each candidate pose using a **Pickability Score** derived from the operational insights of RQ3:

- **Surface Normal Alignment (SNE):** Angular deviation between the predicted surface normal and the world-up vector, critical for a vertical suction approach.
- **Volumetric Support:** The score is weighted by the nominal face area of the object, prioritizing larger, more stable surfaces for vacuum creation.

By selecting the view with the highest functional score rather than the highest visibility, the robot successfully executes picks even when the target remains 40% occluded across most viewpoints.

System Architecture: Heterogeneous ROS2 Bridge

The demonstration is built upon a modular architecture that decouples perception from simulation. Any6D operates as a **Perception-as-a-Service** node on a Linux environment, communicating with the Isaac Sim instance via a custom ROS2 bridge:

- **Asynchronous Inference Loop:** Isaac Sim publishes synchronized RGB-D and segmentation mask streams.
- **Calibration at Runtime:** The ROS2 bridge applies the coordinate transformations identified in RQ1, mapping the latent operational frame of the reconstructor into the robot's world frame in real-time.
- **Operational Precision:** A UR10 Pick-and-Place controller utilizes the selected best-view pose to plan approach trajectories, achieving sub-centimeter contact precision in cluttered bin scenarios.

The demonstration serves as empirical proof of the core thesis: translational stability provided by Any6D is sufficient for autonomous manipulation when guided by a task-aware multi-view policy. Successful suction picks across the YCB representative set—despite high canonical rotational errors in individual views—validate that model-free frameworks are ready for deployment in non-deterministic industrial environments.

6 Conclusion

This report presented a comprehensive evaluation and integration framework for model-free 6D pose estimation using Any6D within a high-fidelity synthetic environment. By systematically addressing coordinate frame alignment and environmental perturbations, the study provides a rigorous foundation for deploying general-purpose perception systems in vision-guided robotic manipulation.

Final Remarks. The results confirm that the transition toward **model-free, category-agnostic perception** represents a viable path for open-world robotics. Any6D's ability to provide stable spatial mass centers and functionally useful contact surfaces without instance-specific training offers a scalable alternative to traditional model-based approaches. By combining advanced 3D reconstruction models with robust pose refinement and task-aware evaluation, it is possible to bridge the gap between simulation-based research and reliable robotic deployment. Future developments will focus on real-time adaptation and multi-modal feedback to further enhance the resilience of autonomous systems in truly novel environments.

References

[1] G. Wang, F. Manhardt, F. Tombari, and X. Ji, "GDR-Net: Geometry-guided direct regression network for monocular 6d object pose estimation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

[2] T. Lee, B. Wen, M. Kang, G. Kang, I. S. Kweon, and K.-J. Yoon, "Any6d: Model-free 6d pose estimation of novel objects," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11633–11643, 2025.

[3] J. Xu, W. Cheng, Y. Gao, X. Wang, S. Gao, and Y. Shan, "InstantMesh: Efficient 3d mesh generation from a single image with sparse-view large reconstruction models," *arXiv preprint arXiv:2404.07191*, 2024.

[4] B. Wen, W. Yang, J. Kautz, and S. Birchfield, "FoundationPose: Unified 6d pose estimation and tracking of novel objects," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[5] Y. Xiang, T. Schmidt, V. Narayanan, and D. Fox, "PoseCNN: A convolutional neural network for 6d object pose estimation in cluttered scenes," in *Robotics: Science and Systems (RSS)*, 2018.

[6] E. Brachmann, A. Krull, F. Michel, S. Gumhold, J. Shotton, and C. Rother, "Learning 6d object pose estimation using 3d object coordinates," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2014.

[7] H. Wang, S. Sridhar, J. Huang, J. Valentin, S. Song, and L. J. Guibas, "Normalized object coordinate space for category-level 6d object pose and size estimation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

[8] B. Calli, A. Singh, A. Walsman, S. Srinivasa, P. Abbeel, and A. M. Dollar, "The YCB object and model set: Towards common benchmarks for manipulation research," in *Proceedings of the International Conference on Advanced Robotics (ICAR)*, 2015.

[9] T. Hodaň, F. Michel, E. Brachmann, W. Kehl, A. Glent-Buch, D. Kraft, B. Drost, J. Vidal, I. Ihrke, X. Zabulis, *et al.*, "BOP: Benchmark for 6d object pose estimation," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018.

[10] R. Szeliski, *Computer vision: algorithms and applications*. Springer Nature, 2022.

[11] B. K. P. Horn, "Closed-form solution of absolute orientation using unit quaternions," *Journal of the Optical Society of America A*, vol. 4, no. 4, pp. 629–642, 1987.